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Physics-data hybrid-driven neural network for radiation-pattern analysis of optical nanostructures

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ABSTRACT

Far-field radiation pattern is a pivotal metric for characterizing the electromagnetic response of dielectric nanostructures. Traditional full-wave numerical methods, including the finite-difference time-domain (FDTD) method, suffer from high computational cost, whereas pure data-driven deep learning models with limited physical interpretability and fixed angular resolutions require large-scale labeled datasets. Herein, we propose a physics-data hybrid-driven neural network (PDHDNN) that integrates exact multipole expansion, including the electric dipole, magnetic dipole, electric quadrupole, and magnetic quadrupole, as a physical prior for fast and accurate radiation-pattern prediction of silicon nanostructures in the optical near-infrared band. The PDHDNN maps structural parameters and frequencies to low-dimensional multipole moments through neural networks and synthesizes the far-field radiation pattern through a physics-constrained multipole-superposition layer. The predicted total scattering cross sections and radiation patterns agree well with FDTD results, showing low relative errors and high normalized correlation coefficients at resonance peaks. At 1° angular resolution, the PDHDNN reduces data storage by 99.83% and shortens training time by 92.6% compared with a conventional pure data-driven network, while enabling arbitrary-resolution radiation-pattern prediction. This work provides an interpretable, low-cost, and efficient surrogate model for nanophotonic radiation analysis and shows potential for the intelligent design of all-dielectric nanophotonic devices.

I. INTRODUCTION

Nanophotonics enables the manipulation of light–matter interactions at subwavelength scales,[1–5] and the far-field radiation pattern represents a core metric for characterizing the electromagnetic response of dielectric nanostructures. Accurate and efficient prediction of radiation patterns is essential for the design of high-performance nanophotonic devices, including nanoantennas, metasurfaces, biosensors, and low-loss photonic components. Conventional full-wave numerical methods, represented by the finite-difference time-domain (FDTD) and finite element method (FEM), provide reliable solutions to Maxwell's equations but incur prohibitive computational costs and heavy memory consumption as the structural complexity and angular resolution increase, severely limiting large-scale simulation and inverse design applications.[6–9]

In recent years, data-driven deep learning has emerged as an acceleration tool for electromagnetic response prediction,[10–13] but pure data-driven neural networks face inherent limitations in radiation-pattern analysis: they require massive high-dimensional labeled datasets and produce black-box outputs lacking physical interpretability and can only generate radiation patterns at a fixed angular resolution after training. Physics-informed and physics-data hybrid-driven neural networks have been developed to address these issues by embedding physical constraints into model training, balancing accuracy, efficiency, and interpretability.[14–18] Among various physical priors, multipole expansion is a fundamental and suitable framework for nanoscale radiation analysis, as it decomposes the complex far-field radiation into a finite set of physically meaningful multipole moments (electric dipole, magnetic dipole, electric quadrupole, magnetic quadrupole).[19–22]

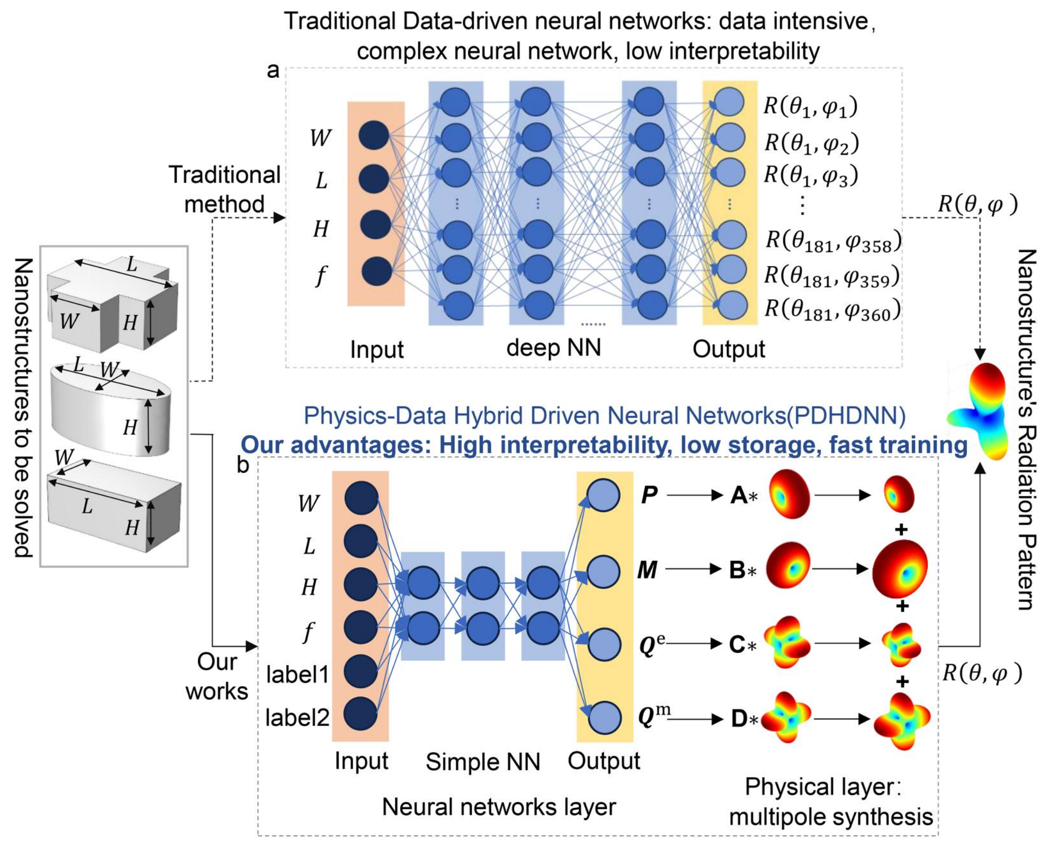

FIG. 1. Schematic of the PDHDNN. (a) Traditional data-driven method with a deep neural network and high-dimensional fixed-resolution outputs. (b) Proposed PDHDNN with a compact neural-network layer and a physics-based multipole-synthesis layer. The neural-network layer predicts multipole moments, and the physical layer combines these moments to reconstruct the radiation pattern of the target nanostructure.

In modern nanophotonics, the optical radiation of all-dielectric nanostructures can be dominated by the interference of electric multipole moments and magnetic multipole moments rather than a single dipole.[23] By engineering the relative amplitudes and phases of these multipoles, one can realize exotic phenomena, including unidirectional scattering, zero backscattering (Kerker effect),[24,25] nonradiating anapole states,[26–28] Fano resonances,[29,30] superscattering,[31,32] and cloaking.[33,34] Within this context, the multipole expansion under the long-wavelength approximation has been widely adopted to describe and interpret light–matter interactions in nanophotonics. Owing to its intuitive physical picture, it further enables the design of a variety of high-performance devices, including directional nanoantennas,[35] metasurfaces,[36,37] nonlinear optical converters,[38] nanolasers,[39] biosensors,[40] and low-loss metamaterials.[41,42]

However, traditional multipole calculations based on the long-wavelength approximation can introduce significant errors when the structural dimensions become comparable to the operating wavelength.[43] To overcome the limitation, recent studies have established exact multipole expansion formalisms applicable to arbitrary particle sizes and shapes, which enable accurate quantitative reconstruction of far-field radiation patterns and scattering cross sections.[44–46] Inspired by the exact multipole expressions, we propose a physics-data hybrid-driven neural network (PDHDNN) for fast and accurate prediction of the radiation patterns of silicon nanostructures at optical frequencies.

Unlike pure data-driven models that directly learn high-dimensional far-field data, the PDHDNN maps structural parameters and frequencies to low-dimensional multipole moments via a neural network and then uses a hard-coded physical layer based on exact multipole superposition to synthesize radiation patterns at arbitrary angular resolutions. This design significantly reduces data storage and training costs, offers full physical interpretability, and maintains accuracy consistent with full-wave simulations. As shown in Fig. 1(a), pure data-driven methods take structural parameters as input and directly predict field values at fixed spatial resolutions, requiring deep, complex networks with high data demand and poor interpretability. In contrast, the PDHDNN in Fig. 1(b) leverages linear superposition of dipole moments and quadrupole moments from multipole decomposition, greatly simplifying the network architecture and enabling far-field prediction at any spatial resolution. The proposed framework provides an efficient, interpretable, resolution-independent surrogate model for nanophotonic radiation analysis and can be extended to broadband responses, multi-material systems, and inverse design.

II. METHOD

This section describes the PDHDNN framework, including the physical foundation based on the multipole expansion method (MEM), FDTD-based dataset generation, neural-network architecture, and training strategy. The central idea is to combine electromagnetic prior knowledge with data-driven learning so that the surrogate model remains efficient while preserving a clear physical connection to far-field radiation.

A. Physical foundation: Exact multipole expansion for far-field radiation

MEM serves as the physical bridge between the structural parameters of a nanostructure and its far-field radiation behavior, while avoiding the direct high-dimensional parameter-to-field mapping required by traditional data-driven approaches. The far-field response of the dielectric nanostructures considered here is described by the exact multipole expansion formalism, which reduces the errors introduced by the conventional long-wavelength approximation for electrically finite particles.[44–46]

For the subwavelength silicon nanostructures studied in the near-infrared regime, the scattered field is dominated by four low-order multipole terms: the electric dipole $\boldsymbol{P}$, magnetic dipole $\boldsymbol{M}$, electric quadrupole $\boldsymbol{Q}^{\mathrm{e}}$, and magnetic quadrupole $\boldsymbol{Q}^{\mathrm{m}}$. These multipole moments are obtained from the induced current density $\boldsymbol{J}(\boldsymbol{r})$ through volume integration.[47] The total scattering cross section can be written as

$$C_{\mathrm{sca}}^{\mathrm{total}} \cong \frac{k^4}{6\pi\varepsilon_0^2 E_0^2}\left[|\boldsymbol{P}|^2 + \left|\frac{\boldsymbol{M}}{c}\right|^2 + \frac{1}{120}\left(|\boldsymbol{Q}^{\mathrm{e}}|^2 + \left|\frac{k\boldsymbol{Q}^{\mathrm{m}}}{c}\right|^2\right)\right]. \tag{1}$$

The far-field scattered electric field is synthesized by the coherent superposition of the radiated fields associated with the retained multipole moments,[48]

$$\boldsymbol{E}_{\mathrm{sca}}(\boldsymbol{n}) \cong \frac{k_0^2}{4\pi\varepsilon_0}\frac{\mathrm{e}^{ik_0 r}}{r}\left\{[\boldsymbol{n}\times(\boldsymbol{P}\times\boldsymbol{n})] - \frac{1}{c}(\boldsymbol{n}\times\boldsymbol{M}) + \frac{ik_0}{6}[\boldsymbol{n}\times(\boldsymbol{n}\times(\boldsymbol{Q}^{\mathrm{e}}\cdot\boldsymbol{n}))] + \frac{ik_0}{6c}[\boldsymbol{n}\times(\boldsymbol{Q}^{\mathrm{m}}\cdot\boldsymbol{n})]\right\}, \tag{2}$$

where r is the radial distance in spherical coordinates, k_0 is the free-space wavenumber, ε_0 is the vacuum permittivity, c is the speed of light, and $\boldsymbol{n}$ is the unit vector along the far-field observation direction. This expression forms the fixed physical layer of the PDHDNN. Instead of requiring labeled far-field values at every observation angle, the proposed method stores and learns the corresponding low-dimensional multipole representation and reconstructs the radiation pattern through multipole synthesis.

A practical criterion for determining whether the four-multipole truncation remains adequate is to compare the truncated multipole reconstruction with the full-wave result. For example, we define the scattering contribution factor,

$$\eta_2 = \frac{C_{\mathrm{sca}}^{P} + C_{\mathrm{sca}}^{M} + C_{\mathrm{sca}}^{Q^e} + C_{\mathrm{sca}}^{Q^m}}{C_{\mathrm{sca}}^{\mathrm{FDTD}}}, \tag{3}$$

where $C_{\mathrm{sca}}^{\mathrm{FDTD}}$ is the full-wave FDTD scattering cross section and η_2 quantifies the fraction of total scattering captured by the retained dipole and quadrupole terms. When η_2 remains close to unity (e.g., >0.95) and the reconstructed radiation pattern maintains a high normalized correlation coefficient with the full-wave result, the four-multipole physical layer is sufficient. Otherwise, octupole and higher-order multipoles should be incorporated into the physical synthesis layer.

B. Nanostructure library and FDTD dataset generation

A diversified nanostructure library is constructed using three representative silicon geometries: cuboid, elliptical cylinder, and cross-shaped prism, as shown in Fig. 2(a). These geometries are representative of all-dielectric meta-atoms rather than isolated arbitrary shapes.[22,49] The cuboid provides a simple anisotropic scatterer for tuning electric and magnetic Mie-type resonances through its aspect ratio. The elliptical cylinder provides a smooth and polarization-sensitive geometry, which is useful for controlling multipolar interference and directional scattering.[24,25] The cross-shaped prism introduces more complex current paths and multi-axis geometric features, making it relevant to multi-resonant and polarization-tailored metasurface elements.[36,37] In the investigated near-infrared frequency range, these silicon nanostructures can serve as elementary resonant units for dielectric nanoantennas, metasurfaces, optical filters, and refractive-index sensing platforms.[35,40] Therefore, fast prediction of their scattering spectra and radiation patterns is directly useful for geometry screening and inverse design.

All structures are parameterized by three geometric variables, L, W, and H, as shown in the left panel of Fig. 1. The parameter ranges are $W = 330 - 390$ nm, $L = 680 - 740$ nm, and $H = 280 - 340$ nm, with a step size of 10 nm for each parameter, resulting in $7 \times 7 \times 7 = 343$ structural configurations. The configuration $W = 350$ nm, $L = 700$ nm, and $H = 300$ nm is reserved as an independent sample for blind testing. For each nanostructure type, the FDTD dataset was generated over a geometry-dependent frequency range with 31 uniformly sampled frequency points. Specifically, the elliptical-cylinder samples cover 242–312 THz, while the cuboid and cross-shaped prism samples cover 182–258 and 210–260 THz, respectively. These frequency ranges were selected to include the dominant resonant features of the corresponding nanostructures and were used consistently for neural-network training, validation, and testing.

FDTD simulations were performed to generate the parametric dataset. A plane wave with an amplitude of 1 V/m was incident along the negative z-direction with x polarization. Each nanostructure was placed at the origin of a three-dimensional cubic solution domain with a side length of 1000 nm, and the domain was terminated by perfectly matched layers to suppress boundary reflections. Structured hexahedral grids with a uniform step size of $\Delta x = \Delta y = \Delta z = 20$ nm were adopted. For each structural

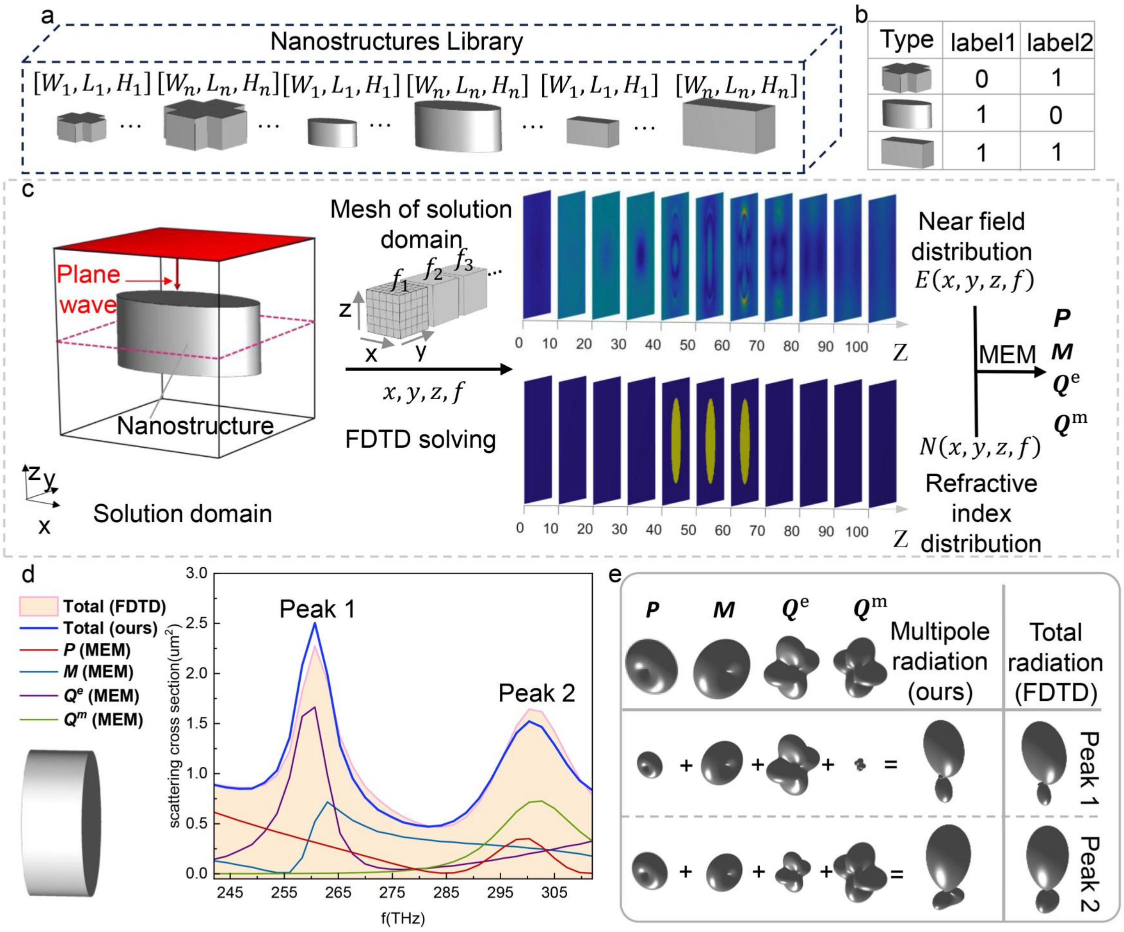

FIG. 2. (a) Illustration of the nanostructures library. (b) Relationship between the nanostructures' type and its label number. (c) Workflow of FDTD-based scattering analysis for resonant dielectric nanostructures and multipole moments decomposition. (d) Calculated scattering cross sections from individual multipole contributions and their total, compared with FDTD results under $W = 350$ nm, $L = 700$ nm, and $H = 300$ nm. (e) The total far-field radiation pattern, reconstructed from the linear superposition of multipole moments derived from near-field data, is compared with the corresponding result obtained from FDTD simulations.

configuration and frequency, the three-dimensional near-field distribution $\boldsymbol{E}(x, y, z, f)$ and the complex refractive-index distribution $n(x, y, z)$ were recorded using Ansys Lumerical FDTD, as illustrated in Fig. 2(c). The nanostructures were modeled as dispersive silicon dielectric structures, and the frequency-dependent complex refractive index was included according to optical-constant data.[50]

C. Architecture of the physics-data hybrid-driven neural network (PDHDNN)

The PDHDNN consists of two functionally distinct modules: a neural-network layer for multipole-moment prediction and a fixed physical layer for multipole synthesis, as shown in Fig. 1(b). This architecture avoids a direct mapping from structural parameters to high-dimensional far-field data and instead learns a compact physical representation. Each output of the neural network corresponds to a physically meaningful multipole moment, which improves interpretability and provides useful information for structural optimization, such as tuning W and L to modulate resonant multipole responses.

The architecture of the PDHDNN is shown in Fig. 3(a). Four independent backpropagation neural networks are trained for $\boldsymbol{P}$, $\boldsymbol{M}$, $\boldsymbol{Q}^{e}$, and $\boldsymbol{Q}^{m}$, while sharing the same input format and hyperparameter settings. The input parameters (W, L, H, f) are min–max normalized to [0, 1], and the multipole outputs are standardized to stabilize training. The input layer contains six inputs, namely, W, L, H, f, label1, and label2, where label1 and label2 are categorical identifiers for the nanostructure type. The two binary labels in Fig. 2(b) are used as compact categorical identifiers rather than physical geometric parameters. Compared with a single scalar type index, this representation avoids imposing an artificial ordinal relation among different shape categories. For the three

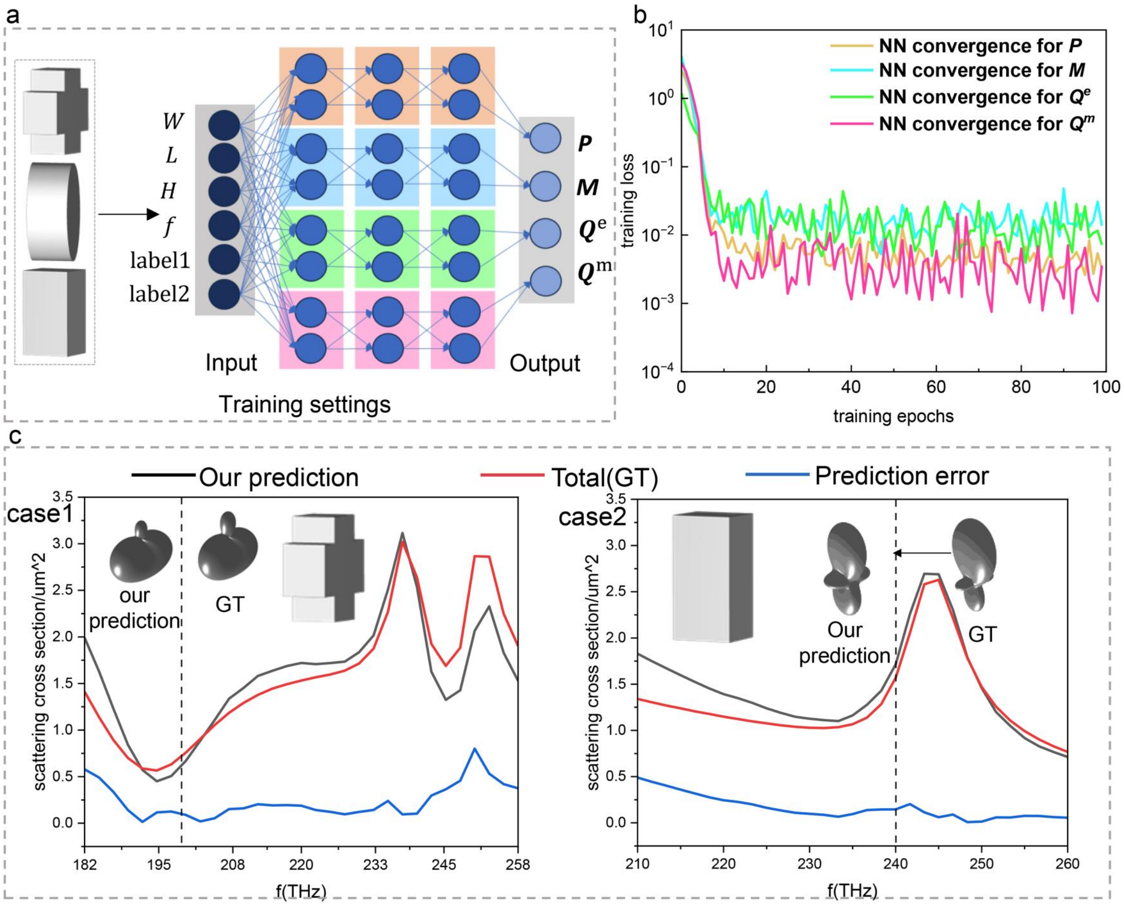

FIG. 3. (a) PDHDNN architecture with structural parameters and type labels as inputs and multipole moments as outputs. (b) Training convergence curves of the four neural networks, plotted against training epochs. (c) Scattering cross sections and radiation patterns predicted by the PDHDNN compared with the FDTD ground truth (GT) for different nanostructure types, both evaluated at $W = 350$ nm, $L = 700$ nm, and $H = 300$ nm.

geometries considered here, two binary labels provide sufficient representation capacity, while one unused binary code remains available for future extension. If a larger or more complex shape library is considered, the label representation can be extended through one-hot encoding, compact binary coding, or learnable geometry embeddings, which are standard categorical representations in neural-network models.[51] For continuously varying topologies, these discrete labels can also be replaced or augmented by geometric descriptors or latent shape representations.

The neural network contains three hidden layers with 30, 25, and 20 neurons, respectively. Tensor symmetry is used to reduce the output dimensionality, namely, $Q^{e}_{\alpha\beta} = Q^{e}_{\beta\alpha}$ and $Q^{m}_{\alpha\beta} = Q^{m}_{\beta\alpha}$, where $\alpha, \beta \in \{x, y, z\}$. In the considered dataset, P_x, M_y, Q^{e}_{zx}, and Q^{m}_{zy} are approximately six orders of magnitude larger than the other corresponding dipole and quadrupole components. These dominant components are, therefore, used as neural-network outputs, while the remaining weak components are neglected in the present implementation.

D. Training strategy and evaluation metrics

The ReLU activation function is applied to the hidden layers to alleviate the gradient-vanishing problem. All networks were trained using the Adam optimizer with an initial learning rate of 10^{-3}. The learning rate was multiplied by a decay factor of $r = 0.95$ every 10 epochs. The maximum number of epochs was set to 1000, and convergence was determined by a validation mean-squared-error threshold of 10^{-4}. The dataset was randomly divided into three subsets: 80% for training, 10% for validation, and 10% for testing.

Two quantitative metrics are used to evaluate prediction performance. The first metric is the relative error, which evaluates the prediction accuracy of the peak position and amplitude of the total scattering cross section,

$$e_r = \frac{|X_{\mathrm{Pred}} - X_{\mathrm{FDTD}}|}{X_{\mathrm{FDTD}}}. \tag{4}$$

The second metric is the normalized correlation coefficient (NCC), which evaluates the consistency between the predicted result and the FDTD reference,

$$\rho = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right]\left[\sum_{i=1}^{n}(Y_i - \bar{Y})^2\right]}}. \quad (5)$$

Here, X_i and Y_i denote the predicted and FDTD reference results, respectively, and $\bar{X}$ and $\bar{Y}$ are their corresponding sample means.

III. RESULTS

A. Validation of the exact multipole expansion physical layer

The multipole synthesis layer is first validated against far-field results obtained from FDTD. Figure 2(d) shows the total scattering cross section of an elliptical cylinder calculated by full-wave FDTD and by MEM summation over the 242–312 THz frequency range corresponding to the elliptical-cylinder dataset. The two curves almost overlap in this frequency range. The main resonance peak is located at 260 THz with a relative error below 8.1%, and the secondary peak is located at 300 THz with a relative error below 6.3%. The electric quadrupole Q^e dominates the main low-frequency resonance, the magnetic dipole M provides the secondary contribution, and P and Q^m further refine the radiation details.

The radiation patterns at the two resonance peaks are compared in Fig. 2(e). The upper part of the panel shows the individual radiation patterns associated with the electric dipole P, magnetic dipole M, electric quadrupole Q^e, and magnetic quadrupole Q^m. The lower part compares the total radiation patterns reconstructed by their coherent superposition with the FDTD reference. For both resonance peaks, the reconstructed patterns agree well with the FDTD results, with NCC values of 0.96 and 0.92, respectively. These results confirm that the retained multipole channels provide an accurate physical layer for the considered electrical-size range.

B. Prediction accuracy of the PDHDNN

The convergence behavior of the four independently trained neural networks is shown in Fig. 3(b). The training loss curves decrease steadily with increasing training epochs. Although the prescribed convergence loss was set to 10^{-4}, subsequent experiments showed that the networks could not fully reach this target. Nevertheless, stopping training after 100 epochs already gave satisfactory performance, with significantly reduced training time. This indicates stable optimization for all retained multipole components. The total scattering cross sections and radiation patterns predicted by the PDHDNN are further compared with FDTD ground truth for different nanostructure types in Fig. 3(c). For both Case 1 and Case 2, the predicted cross sections follow the same spectral trend as the FDTD results, and the main resonance positions are accurately reproduced.

In Case 1, the main peak occurs at 238 THz, with a peak relative error below 5.8%. In Case 2, the main peak is located at 245 THz, with a peak relative error below 8.2%. The radiation patterns predicted at selected frequencies also match the FDTD references closely, yielding NCC values of 0.88 and 0.94 for Case 1 and Case 2, respectively. These results show that the PDHDNN can learn the mapping from structural parameters and frequency to physically meaningful multipole moments and can accurately reconstruct the far-field radiation behavior of different silicon nanostructures.

In Case 2, a slightly increased error is observed near the low-frequency side around 210 THz. This deviation arises mainly because this frequency region is close to the lower boundary of the training range, where the multipole responses contain fewer pronounced spectral features for the neural network to learn. Nevertheless, the dominant resonance position and the overall radiation pattern remain well reproduced, as confirmed by the NCC value. Therefore, the observed low-frequency deviation does not affect the main predictive capability of the proposed framework and can be fixed given more training data.

C. Resolution independence and comparison with the pure data-driven model

A key advantage of the PDHDNN is its resolution-independent prediction capability, as shown in Fig. 4(a). Once the multipole moments are predicted, the far-field radiation pattern can be synthesized at arbitrary angular resolutions by changing the observation direction vector $\boldsymbol{n}$ in the physical layer, without retraining the neural network. In contrast, a conventional pure data-driven neural network directly learns far-field values at predefined angular sampling points, and its output dimension is, therefore, tied to a fixed angular resolution.

To ensure a fair comparison, the conventional pure data-driven neural network was evaluated on the same test set using the same prediction metrics. As summarized in Table I, at 1° angular resolution, the pure data-driven NN achieves peak relative errors of 5.1% and 8.4% for Case 1 and Case 2, respectively, with corresponding radiation-pattern NCC values of 0.90 and 0.92. Under the same setting, the proposed PDHDNN yields peak relative errors of 5.8% and 8.2% and NCC values of 0.88 and 0.94, indicating comparable prediction accuracy at high angular resolution. When the angular resolution is changed to 10°, the pure data-driven NN shows larger peak relative errors of 13.9% and 17.6%, whereas the PDHDNN maintains the same peak relative error level of 5.8% and 8.2%. These results further confirm that the physical multipole-synthesis layer provides better resolution robustness.

The storage requirement and training time are further compared in Figs. 4(b) and 4(c). Since the PDHDNN learns low-dimensional multipole moments and uses the fixed physical layer for radiation-pattern synthesis, its dataset size remains 0.34 GB and its training time remains 37.28 min for different angular resolutions. By contrast, the pure data-driven model requires far-field labels at all angular sampling points, so its storage requirement and training time increase rapidly as the angular resolution becomes finer. At 1° angular resolution, the pure data-driven model requires 273.28 GB of data storage and 567.9 min of training time, whereas the PDHDNN reduces the data storage by 99.83% and the training time by 92.6%. Therefore, the efficiency

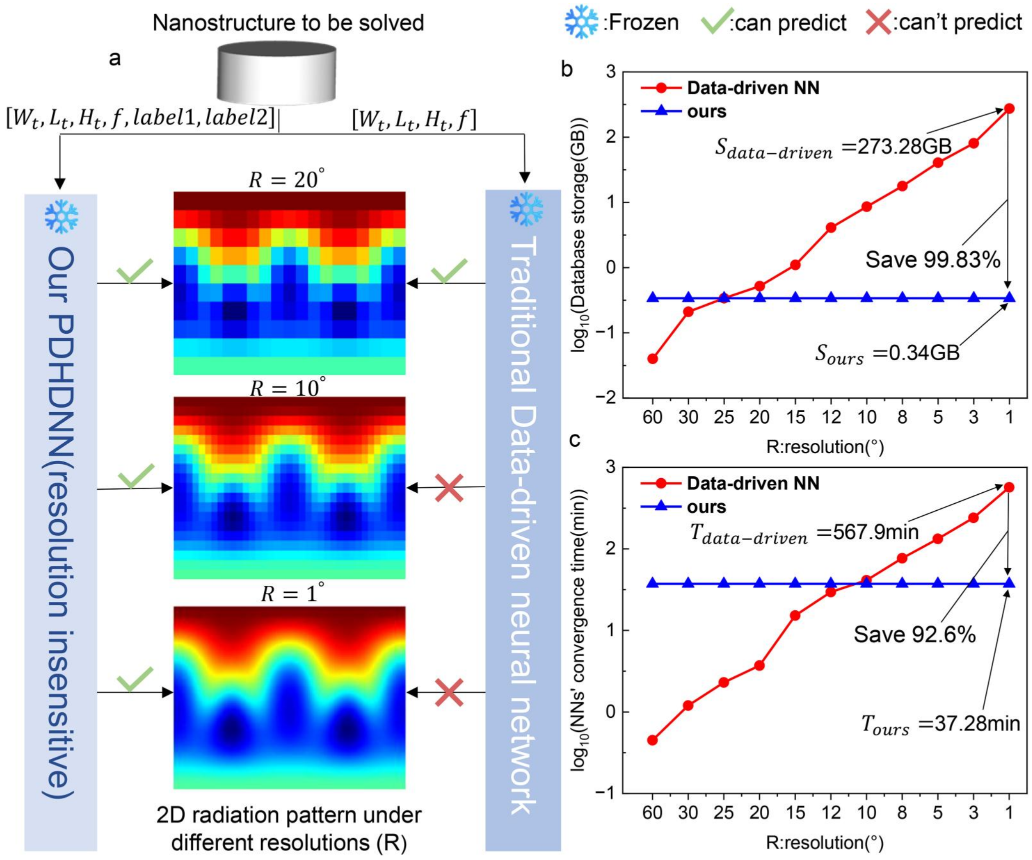

FIG. 4. Resolution-independent prediction and efficiency comparison. (a) Radiation patterns generated by the PDHDNN at different angular resolutions. (b) Database-storage requirement vs angular resolution. (c) Neural-network convergence time vs angular resolution.

gains shown in Figs. 4(b) and 4(c) are achieved without sacrificing the prediction accuracy required for radiation-pattern analysis.

D. Limitations and future outlook

Despite its demonstrated accuracy and efficiency, the present PDHDNN still has several limitations that should be addressed in future works. First, the current implementation retains four dominant multipole terms, namely, the electric dipole, magnetic dipole, electric quadrupole, and magnetic quadrupole. This truncation is sufficient for the subwavelength silicon nanostructures considered in this work because these four terms dominate the scattering spectra and the reconstructed radiation patterns within the investigated frequency range. More generally, the required truncation order is governed by the electrical size of the scatterer. As this size increases, or when the induced current distribution exhibits stronger spatial variation, higher-order resonances can appear and the contribution of octupole or even higher-order multipoles become non-negligible. The proposed framework is not intrinsically limited to four multipoles; the present truncation is an

TABLE I. Prediction accuracy comparison between the proposed PDHDNN and the pure data-driven NN.

Method	Resolution (°)	Peak relative error Case 1 (%)	Peak relative error Case 2 (%)	Pattern NCC Case 1	Pattern NCC Case 2
Proposed PDHDNN	1	5.8	8.2	0.88	0.94
Pure data-driven NN	1	5.1	8.4	0.90	0.92
Proposed PDHDNN	10	5.8	8.2	0.91	0.93
Pure data-driven NN	10	13.9	17.6	0.84	0.82

accuracy-efficiency trade-off validated for the electrical size range studied here. For larger or more complex structures, higher-order multipoles can be incorporated into the physical synthesis layer following the same validation criterion.

Second, the current model is tailored to silicon dielectric nanostructures. Extending the framework to other dielectric materials or metal-dielectric hybrid systems will require incorporating the corresponding dispersive material parameters and recalculating the induced current distribution used for multipole-moment extraction. Third, the present validation is performed under an x-polarized plane wave propagating along the negative z-direction. Future extensions should include arbitrary incident directions and polarization states so that the model can be applied to more general optical scattering and metasurface-design scenarios. These extensions are technically compatible with the proposed PDHDNN architecture: The neural network can be expanded to predict additional multipole moments or include incident-field parameters, while the physical synthesis layer can still enforce electromagnetic radiation laws through multipole superposition.

IV. CONCLUSIONS

This work proposed a PDHDNN for fast and accurate radiation-pattern prediction of silicon nanostructures at optical frequencies. The framework maps frequency, nanostructure type, and structural parameters to physically meaningful multipole moments and then reconstructs the far-field radiation pattern through an exact multipole-synthesis layer. In this way, the model combines the interpretability of multipole physics with the efficiency of neural-network regression. The results show that the PDHDNN generalizes to cuboid, elliptical-cylinder, and cross-shaped-prism silicon nanostructures, accurately reproduces scattering spectra and radiation patterns, and supports arbitrary angular resolutions without retraining.

Compared with a conventional pure data-driven neural network, the PDHDNN substantially reduces data storage and training time while maintaining comparable prediction accuracy. These advantages arise from learning low-dimensional multipole labels rather than dense far-field samples. The proposed framework, therefore, provides an efficient and interpretable surrogate model for nanophotonic radiation analysis and geometry screening. Future work will extend the framework to higher-order multipole moments, broader material systems, arbitrary incidence conditions, and inverse design of nanophotonic devices.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Wiong Wei Wu and Qi Cheng Chen contributed equally to this paper.

Xiong Wei Wu: Conceptualization (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal). **Qi Cheng Chen:** Conceptualization (equal); Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal). **Jian Lin Su:** Formal analysis (equal); Investigation (equal); Validation (equal). **Xuan Zheng:** Formal analysis (equal); Investigation (equal); Validation (equal). **Jun Ming Hou:** Formal analysis (equal); Investigation (equal); Validation (equal). **Yiqian Mao:** Funding acquisition (equal); Validation (equal); Writing – review & editing (equal). **Jian Wei You:** Funding acquisition (equal); Investigation (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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